



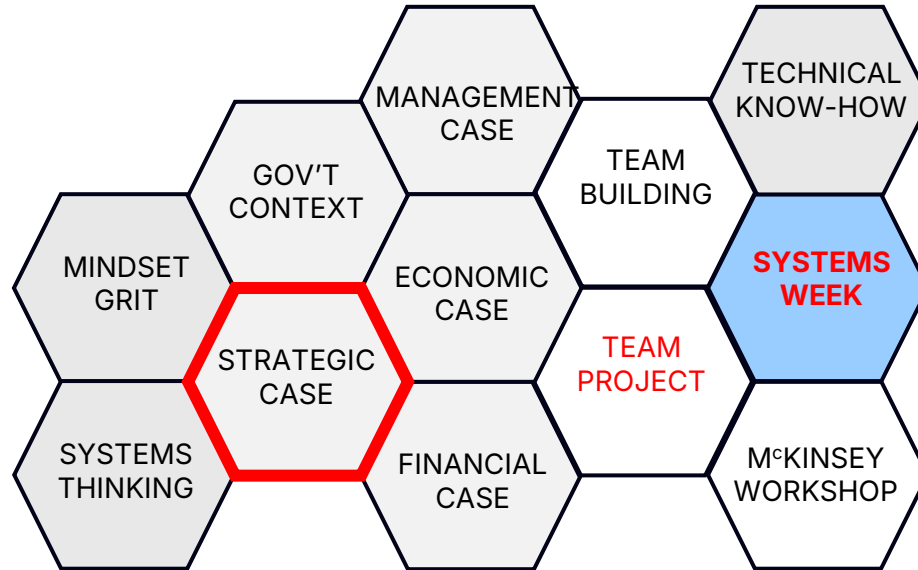
Waipapa
Taumata Rau
**University
of Auckland**

Defining and scoping your wicked problem

Lecture 9

>>> 07 August 2026

>>> Marc lewis and Amanda Di lenno





Outline

- The components of a problem statement
- Framing a systems problem
- Causal loops and systems level problems
- Defining problem spaces
- Identification of the problem statement
- Reminders



Learning outcomes

- Define the components of a problem statement
- Frame System Level Problems using Systems Thinking tools to identify problem statements that address one or multiple aspects of the problem space

Problem Finding

Complex System

Problem Space

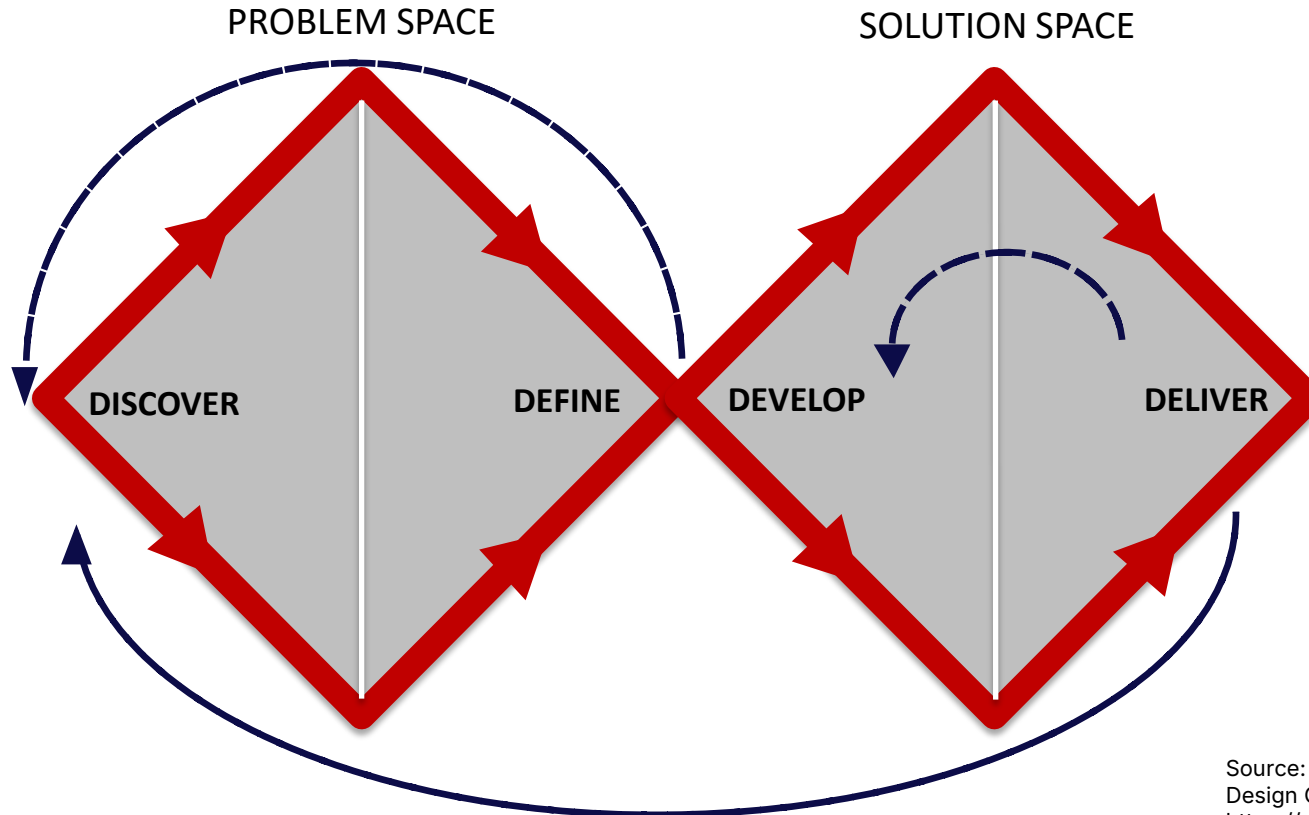
Problem Statement



“There Are No Solutions to Complex Challenges
Only an endless series of adaptive responses”

David Gurteen

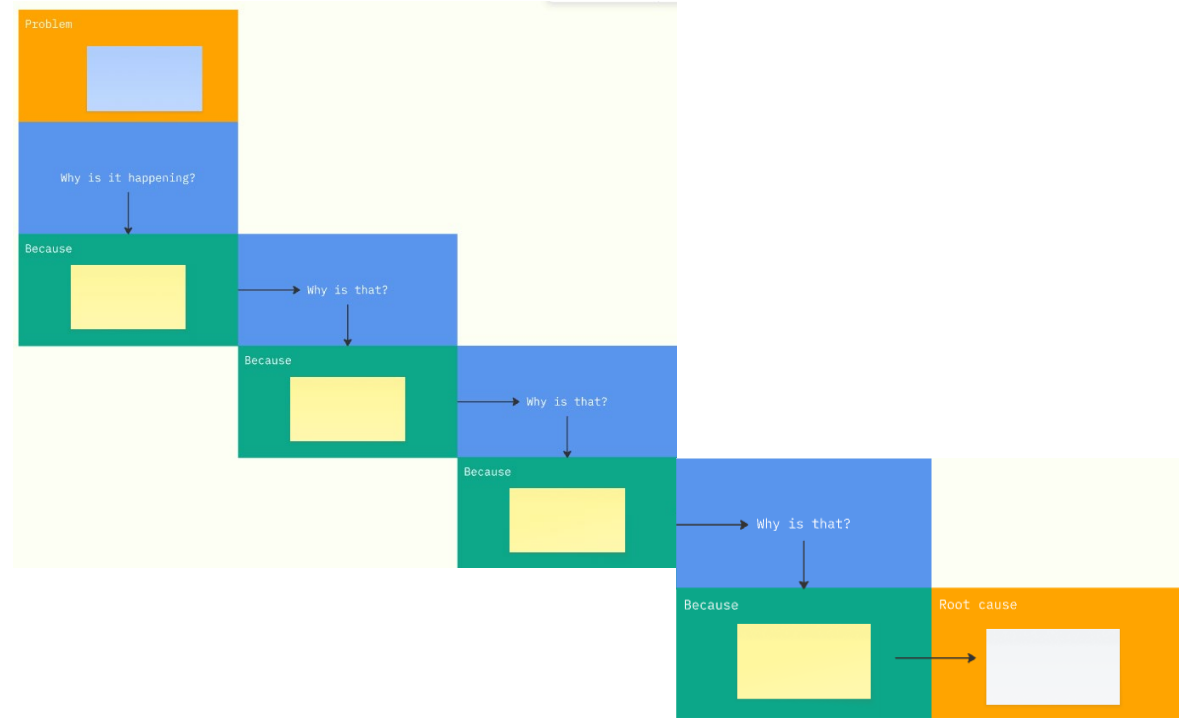
Divergent & Convergent Thinking



Source: Inspired by Double Diamond,
Design Council,
<https://www.designcouncil.org.uk/>

5 whys

- The Five Whys helps identify the root causes of complex problems.
- Complements systems thinking by revealing the variables that drive system behaviour.
- Today, we'll build on this by connecting those variables into a causal loop diagram to identify feedback loops and leverage points.



Rethinking plastics – 2025 systems week

What new innovations can be developed to tackle the plastic problem in New Zealand over the next 10 years?

#rethinkplastic
*Make best practice,
standard practice*

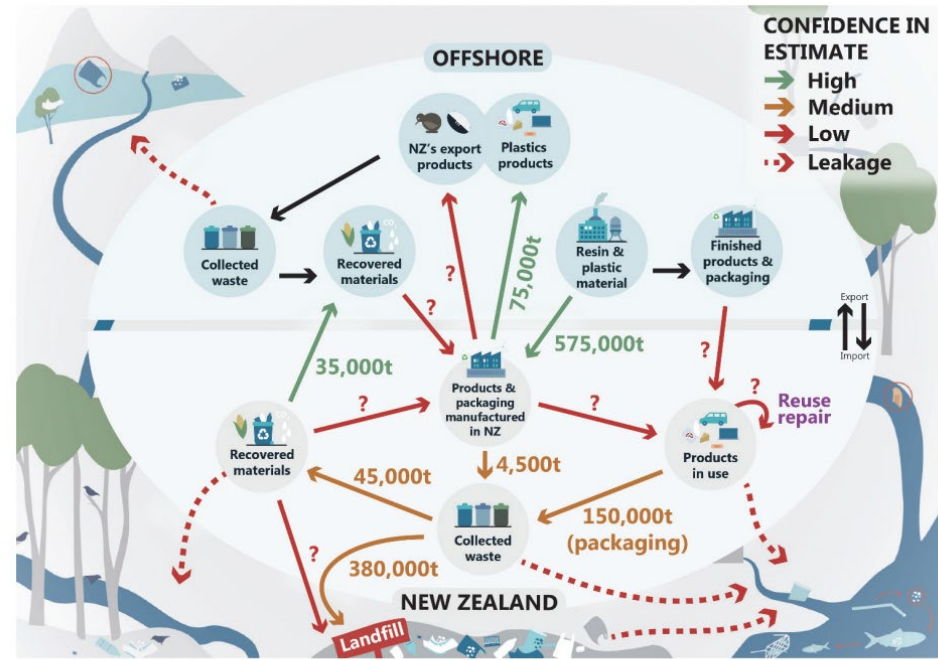
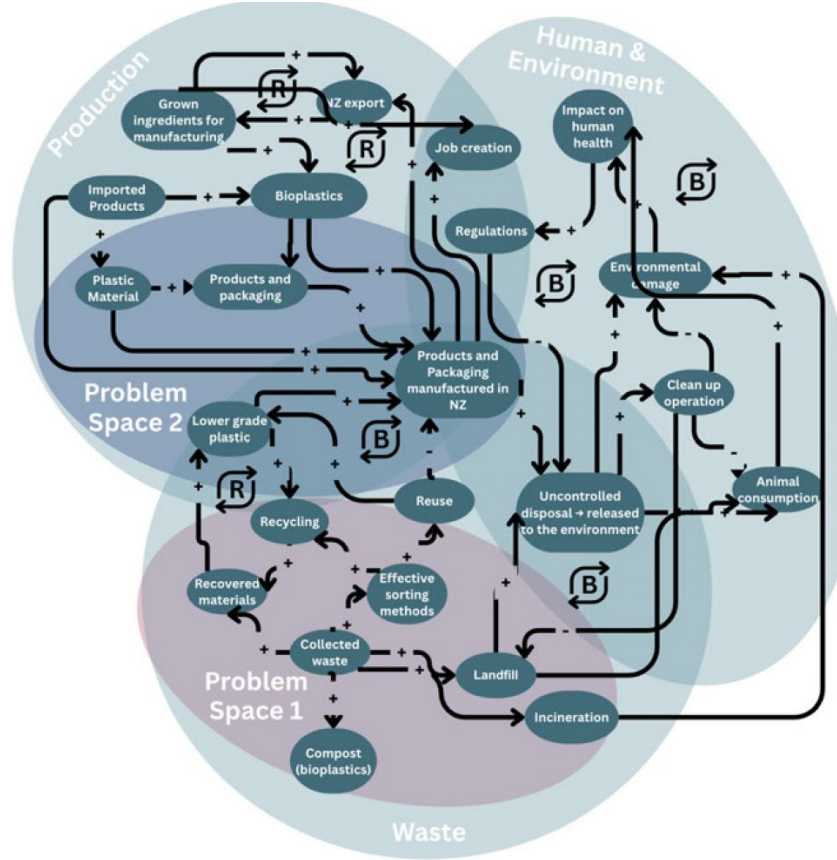


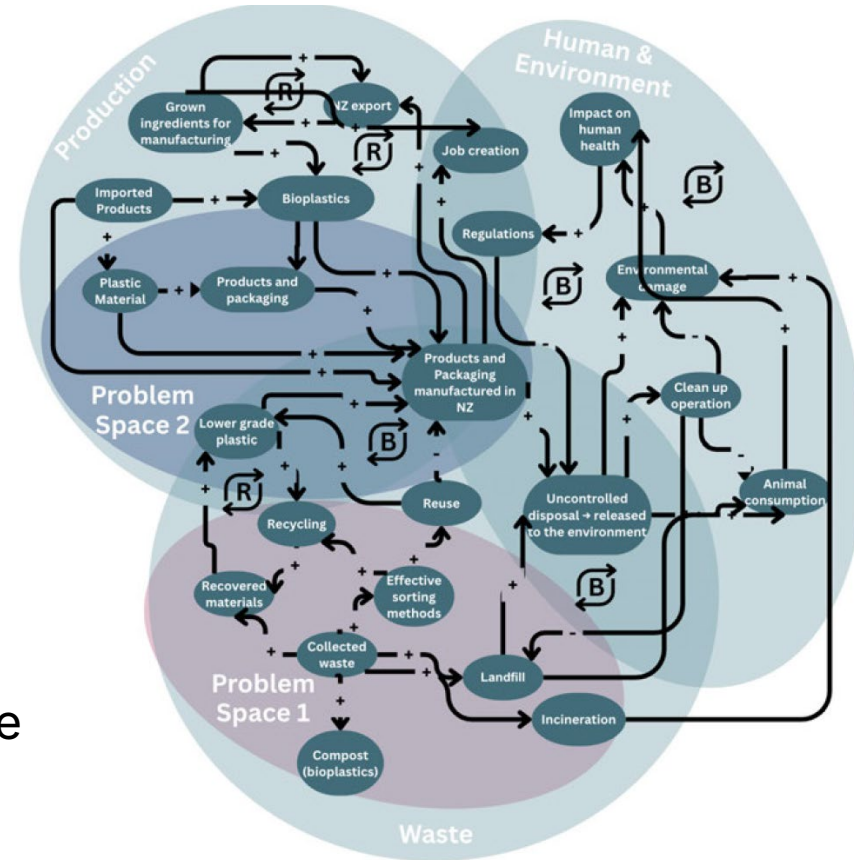
Figure 12 Flows of plastic into and out of Aotearoa New Zealand, including leakage into the environment. Plastic leaking into the environment includes macro and microplastics, and affects land, marine and air environments. It also includes waste that is burned or buried in unregulated landfill. Sources of data are outlined in Table 3.



Further reading: Using mind maps in agricultural systems (see [here](#))

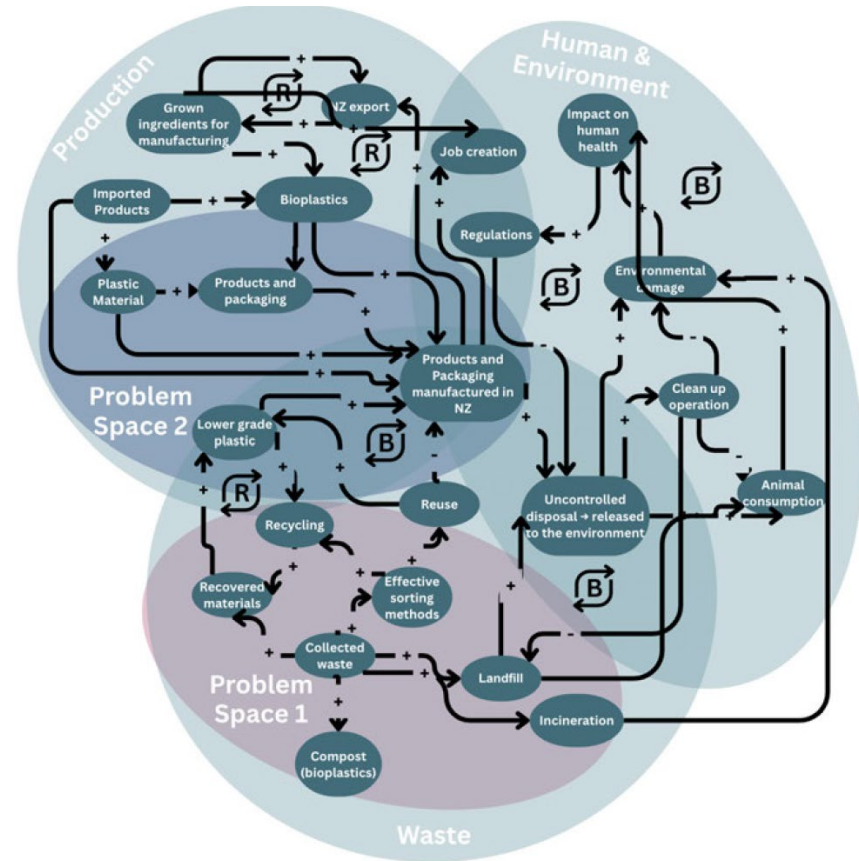
Causal Loop diagram

- **Variables (nodes)** represent the key factors within a system.
- Arrows show **cause and effect**.
- **+** = variables move the **same direction**.
- **-** = variables move **opposite directions**.
- **Reinforcing (R)** = amplifies change.
- **Balancing (B)** = counteracts change.
- **Leverage points** = where interventions have the greatest impact.
- **Problem statements** summarise what the diagram reveals about the system.



Causal Loop diagram

1. **Identify variables** in the system.
2. **Determine relationships** between variables.
3. **Identify feedback loops** that explain why the problem persists.
4. **Identify leverage points** where interventions could have the greatest impact.
5. **Develop a systems problem statement** describing the problem, who is affected, why it exists, and its consequences.

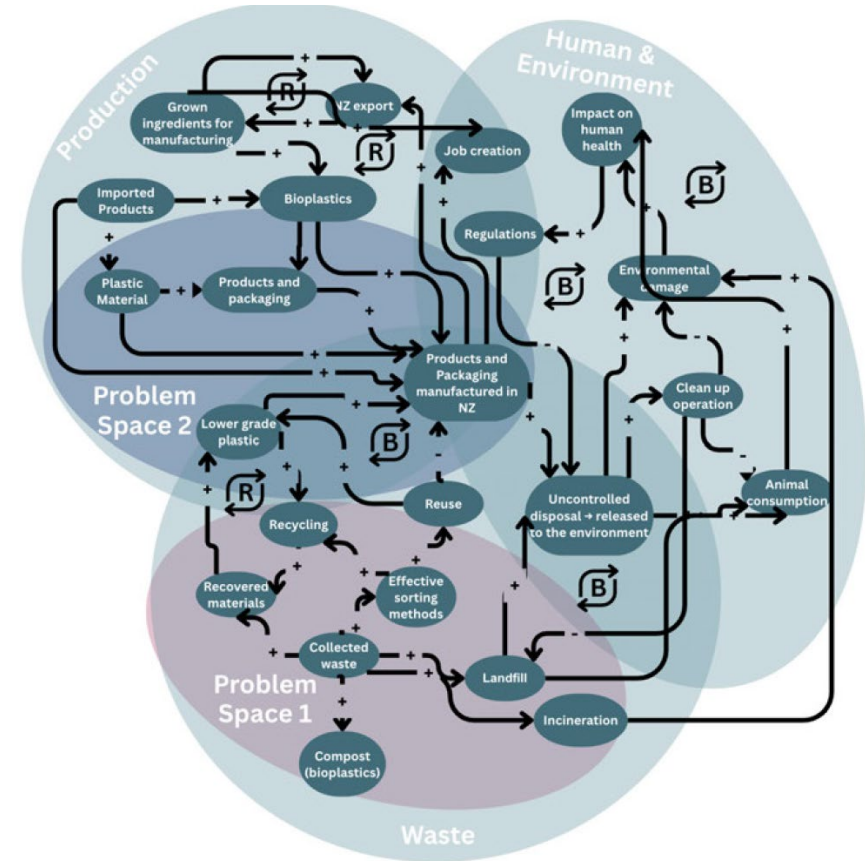


Identifying Leverage points to draw out problem statements

A place in a system where a small change can create a disproportionately large change throughout the entire system.

Look for variables that:

- Are highly connected Influencing multiple feedback loops.
- Affect many other variables.
- Can be realistically changed through an intervention.



B to C Problem Statement – How Might We?

How Might We + **Intended Action**
(as an action verb) + **For** + **Potential User**
(as the subject) + **So That** + **Desired Outcome**

Intended Action

Potential User

Desired Outcome

Problem Statement

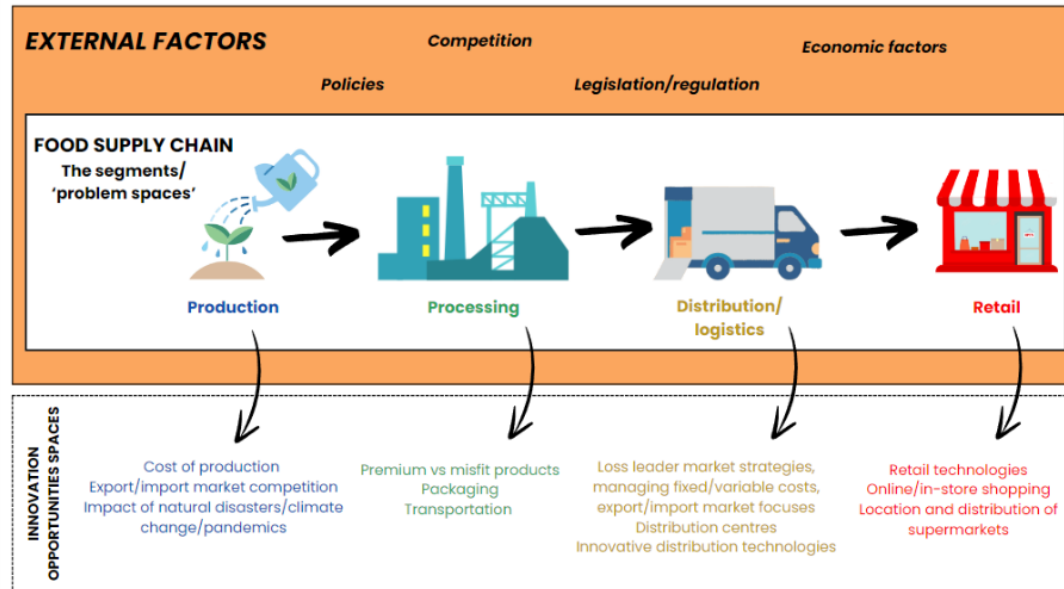
Causal Loops To Problem Statement

The problem is [briefly describe the problem] which affects [who is impacted] because [why the problem exist] leading to [what negative consequences exist].

New Zealand's plastic waste management infrastructure is underdeveloped affecting communities and the environment because limited recycling capacity reduces plastic recovery leading to increased environmental pollution and long-term ecosystem damage.

Activity: Let's build a causal loop for high food prices

High food prices are a complex systems problem with no single cause, emerging from interactions between production, transport, retail, consumers, and government.



PROBLEM FRAMING CANVAS

“If I had an hour to solve a problem I’d spend 55 minutes thinking about the problem and five minutes thinking about solutions”.

Albert Einstein



First-cut problem statement:

Whose problem is it?
(a human view)
What is the need?
Why is this a problem?

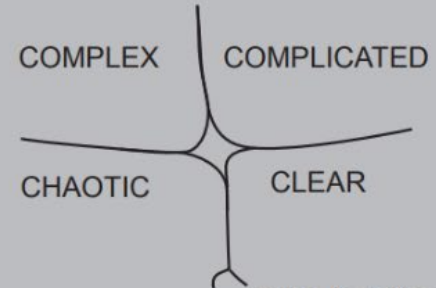
Is there a problem behind the problem?

Any insights from the 5 whys?

Problem framed as an opportunity question - How Might We.....?

Question that has the potential to spark at least 10 ideas about potential responses

What type of problem is this?



Source: thecynefin.co/

Any changes to your first cut problem statement?

Draw out the problem - create a rich picture

What is the story of this problem?
How does the problem 'work'?

Frame the problem in three different ways:

Who cares about the problem enough to act on it? Who has a stake in the problem or its resolution?

Does anyone benefit from the problem as a problem?

What does this picture reveal about how you 'see' the problem?

If you're stuck trying to pull out your problem statements there are many tools to think of

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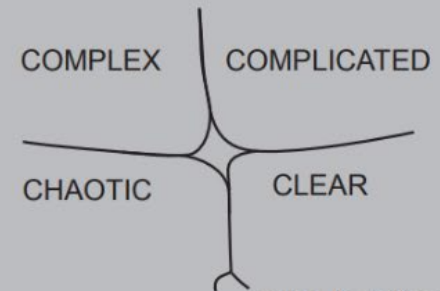
Problem framed as an opportunity question -

How Might We.....?

How Might We _____ *Action*
_____ *What*
for _____ *Stakeholder*
in order to _____ *What change?*

Question that has the potential to spark at least 10 ideas about potential responses

What type of problem is this?

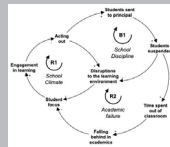


Source: thecynefin.co/

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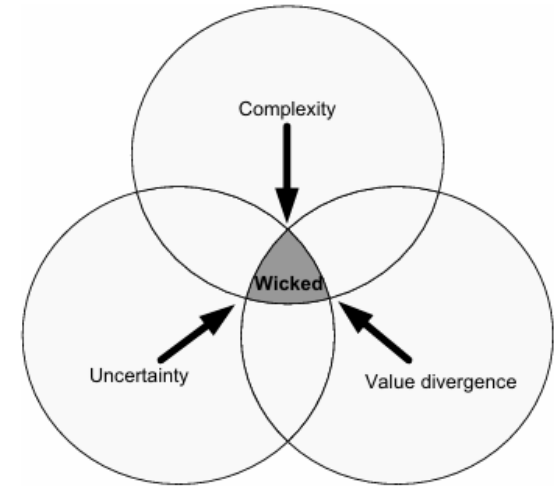
After Class Activity – Complete Your Causal Loop Analysis

Using the causal loop diagram you started in class:

1. Complete your causal loop diagram by identifying the remaining variables affecting food prices.
2. Identify and label at least one reinforcing (R) or balancing (B) feedback loop.
3. Identify one or two leverage points where an intervention could have the greatest system-wide impact.
4. Use these insights to develop a systems problem statement that explains why the problem exists and why it persists.

Summary

- Causal loop diagrams visualise cause-and-effect relationships and system behaviour over time.
- Leverage points are places where small changes can create large system-wide impacts.
- Use causal loop diagrams to develop a clear systems problem statement before considering solutions.





Questions?

Thank you, and see you next
lecture 😊

